

Complements of Operational Research

1.4 Subjective probabilities

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Where do the probabilities come from?

Objective

Estimated from **data**: frequencies observed in the past, under conditions similar to the current ones. E.g.: proportion of months with high demand in the last 5 years.

Subjective

When there are no data (new product, new market), they express the **degree of belief** of the decision maker or of an expert.

Question: the sales director cannot say “the probability of high demand is 0.43”. But he can compare states: “high demand is as likely as medium demand”. Is that enough?

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It is. Comparing two states is easier than assigning an absolute number to each one. The methods in this section turn comparisons into probabilities.

Direct method: relative odds

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- “medium demand is 3 times more likely than low demand”.

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Writing everything in terms of $P(\theta_1)$:

$$P(\theta_2) = 3P(\theta_1), \quad P(\theta_3) = 3P(\theta_1),$$

$$P(\theta_1) + 3P(\theta_1) + 3P(\theta_1) = 1 \iff P(\theta_1) = \frac{1}{7}.$$

Result

$$P = \left(\frac{1}{7}, \frac{3}{7}, \frac{3}{7}\right) \approx (0.14, 0.43, 0.43)$$

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With m states, $m - 1$ comparisons are enough (plus the sum equal to 1).

Opinion changes the decision

	low	medium	high	EV
a_1 : small	40	45	50	$325/7 \approx 46.4$
a_2 : medium	20	60	80	$440/7 \approx 62.9$
a_3 : large	-40	50	120	$470/7 \approx 67.1$
P	1/7	3/7	3/7	

For example,

$$EV_{a_3} = \frac{1}{7}(-40) + \frac{3}{7}(50) + \frac{3}{7}(120) = \frac{470}{7}.$$

With these probabilities the decision becomes a_3 (large).

In 1.3, with (0.3, 0.5, 0.2), the choice was a_2 .

Lesson

The criterion is the same (MEV); what changed is the **information**. A director who is optimistic about demand leads to a larger investment — and he is the one who has to take responsibility for that opinion.

Good practice: record the decision maker's answers; if the decision is discussed later, it is known where the numbers came from.

What if the decision maker gives too many comparisons?

Suppose the decision maker adds: “high is 2 times more likely than low”.

But the first two answers imply

$$\frac{P(\theta_3)}{P(\theta_1)} = \frac{P(\theta_3)}{P(\theta_2)} \cdot \frac{P(\theta_2)}{P(\theta_1)} = 1 \times 3 = 3 \neq 2.$$

The three answers are **inconsistent**: there are no probabilities that satisfy all of them.

Consistency

If $\frac{P_i}{P_k} = a_{ik}$ and $\frac{P_k}{P_j} = a_{kj}$, then we must have
$$\frac{P_i}{P_j} = a_{ik} a_{kj}.$$

In practice, the inconsistency is shown to the decision maker, who is asked to revise one of the answers.

We will meet this idea again in ch. 8 (AHP): pairwise comparisons and a measure of consistency.

Indirect methods

When the decision maker finds it hard to give probability ratios, the probabilities are obtained indirectly:

Method	Idea
I. Equivalent urn	A bet on the event is compared with drawing a ball from an urn of known composition; the composition is adjusted until the decision maker is indifferent between the two bets.
II. Lottery	The same, but comparing with a lottery that pays the same prize with probability p ; p is adjusted until indifference.
III. Histogram smoothing	Uses the histogram of values observed in the past, smoothed; assumes conditions not very different from the current ones.
IV. Quantiles	The decision maker is asked for the values that divide the distribution (median, quartiles) of a continuous quantity.
V. Relative heights	The decision maker draws the relative height (likelihood) of each value or interval; it is normalised to sum to 1.

Methods I and II require the decision maker to compare bets carefully: the answers depend heavily on how the questions are asked.

Example — relative odds between 3 states

The decision maker is asked for the relative odds between the three states of nature, and answers:

“the relative odds of state θ_1 with respect to θ_2 are 3 to 1” and “the relative odds of state θ_2 with respect to θ_3 are 2 to 1”.

Find $P(\theta_1)$, $P(\theta_2)$ and $P(\theta_3)$.

Exercise

With the probabilities from the example, which action should be chosen in the table of Example 3 (investment) by maximisation of the expected value? And if the decision maker had said “1 to 3” and “1 to 2”?

In Exercise sheet 1: Exercises 6, 8 and 9 (tests of 2019, 2024 and 2025).

KEY TAKEAWAYS — 1.4

1. Without data, the probabilities can come from the decision maker's **opinion**: subjective probabilities.
2. **Direct method**: with m states, $m - 1$ relative odds and $\sum_j P(\theta_j) = 1$ form a linear system.
3. The answers must be **consistent**: $a_{ij} = a_{ik} a_{kj}$.
4. **Indirect methods**: equivalent urn, lottery, histogram smoothing, quantiles, relative heights.
5. Once the probabilities are obtained, the criteria of 1.3 (MEV, EOL) are applied — and the decision may change with the opinion.

End of chapter 1. Next: 2 Planning models (CPM, PERT, Gantt).

References and resources

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Decision Analysis.
- Clemen, R. T., Reilly, T., *Making Hard Decisions with DecisionTools*, 3rd ed., Cengage, 2014 — assessment of subjective probabilities.

Resources

- Exercise sheet 1 — Exercises 6, 8 and 9.